ASR Performance Report (VPB Small Dataset)

VPBank AI Center (AIC)

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Introduction and Context

- **VPBank Overview**: A leading bank in Vietnam, pioneering AI-driven financial solutions for digital transformation.
- **AI Center (AIC)**: Dedicated to developing cutting-edge AI technologies for banking operations.
- **Debt Collection Division (DCD)**: Manages 250k customers with 25k daily calls, facing challenges in speech-to-text (STT) due to diverse user voices.
- **Collaboration with AWS**: Utilized $14,000 in GPU credits for training and fine-tuning ASR models.
- **VPB Small Dataset**: 10 hours of sampled call data, split into 5 segments for processing and analysis.
- **Goal**: Build robust, scalable STT models to enhance automated debt collection and customer interaction.

Technical Context

- **Chunkformer (closed-source, current in use):**
  - WER 23-27%, performs well but **cannot be fine-tuned**.
- **Fast-Conformer (open-source, pre-trained on VietSpeech):**
  - WER 35-39%, underperforms.
- **Parakeet (open-source, pre-trained and fine-tuned):**
  - Slightly better and more stable performance than Fast-Conformer (see results).
- **Approach:** Two-step fine-tuning:
  - v1: Pseudo-labeling.
  - v2: Pseudo-labels → VPB training data.
- **Utilization of AWS Credits:** $14,000 in GPU credits used for training and fine-tuning both Fast-Conformer and Parakeet models on AWS infrastructure.

WER Results (Fast-Conformer, VPB Small Dataset)

Dataset	Chunkformer	v0 (VietSpeech)	v1 (Pseudo)	v2 (Pseudo→VPB)	Notes
standard_test_2	0.2499	0.3546	0.3040	0.2420	70% overlap
standard_test	0.1613	0.3378	0.2582	**0.4294**	Small (29 samples)
next_day_test_debug	0.2076	0.3414	0.2687	0.2649	Independent test
vpb_right2_train	0.2420	0.3633	0.2956	0.2456	Training set
vpb_right2_valid	0.2696	0.3902	0.3240	0.2821	Validation set

- Note: Metrics represent performance across 5 segments of VPB Small Dataset (10 hours).

WER/CER Results (Parakeet, VPB Small Dataset)

Dataset	VietSpeech (v0)	Pseudo-labeling (v1)	Pseudo→VPB (v2)
standard_test_2	33.55% / 23.05%	28.29% / 21.87%	23.26% / 17.29%
standard_test	27.92% / 18.01%	21.70% / 15.32%	21.00% / 13.96%
next_day_test_debug	31.24% / 20.29%	25.49% / 18.32%	21.78% / 15.69%
vpb_right2_train	33.65% / 23.36%	27.66% / 21.29%	22.99% / 16.91%
vpb_right2_valid	36.68% / 25.79%	31.33% / 25.05%	26.59% / 20.44%

- Note: Parakeet shows slightly better and more stable WER/CER than Fast-Conformer across all sets (10 hours).

Analysis

- **Fast-Conformer (Pre-trained)**: WER 0.35–0.39, significantly worse than Chunkformer.
- **Fast-Conformer (Fine-tuning v1)**: Strong improvement (WER reduction of 0.07–0.10).
- **Fast-Conformer (Fine-tuning v2)**: Approaches Chunkformer on **key sets (validation, next_day)**.
- **Parakeet**: Consistently outperforms Fast-Conformer with lower WER/CER and better stability across all datasets.
- **Key Notes**:
  - Low WER on **standard_test_2** due to **data leakage** (70% overlap with training).
  - Low WER on **train** indicates convergence, but **not representative of new user performance**.
- **Current Achievement**: Developed fine-tuned Fast-Conformer and Parakeet models using AWS GPU resources.

Key Insights

- **Core Challenge**: Diverse user voices & noise:
  - Regional accents, speech speed, stuttering, local dialects.
  - Environmental noise, poor call signals.
- **Scale of Operations**: 250k customers in debt collection; 25k new calls daily → Major STT issues due to voice diversity vs. public datasets; even basic audio normalization (mean/std-based) fails.
- **Data Accumulation**: Plan to collect larger datasets (6,000 calls/week, 100 hours) for future labeling and processing (pseudo-labeling, self-supervised, user-based STT, etc.).
- **Model Potential**: Build robust models for unseen data (user voices).
- **Solution**: Manual labeling → Internal gold-standard dataset; continuously update with new user voices.
- **Future**: Collect user voices at loan contract stage → User verification, automated debt collection/reminders via voice-calls.

Value of Data Labeling

- **Essential for Fine-Tuning**: Required to improve and sustain performance.
- **Long-Term Data Asset**:
  - Call analysis, user verification, service quality evaluation.
  - Train NLP models.
  - Develop Callbot (automated voice interactions).
- **One-Time Investment**: Reusable across multiple iterations and applications.
- **Internal Potential**: Enhances model accuracy; builds strategic in-house data for competitive edge.

Urgency and Resource Needs

- **AWS Credits**: \$15,000 (expiring 09/20/2025) $\rightarrow$ \$14,000 already utilized for GPU training.
- **Labeling Deadline**: 09/15/2025.
- **Requirements**: 100 staff, 500 overtime hours (100 million VND).
- **Computing Needs**: Additional GPU resources for ongoing training, scaling to handle larger datasets in the future.
- **Investment Alignment**: Coordinate with AWS, DCD, and AIC for further computing + data investments.

Conclusion

- **Without Labeling**: Model stalls at 70-75% accuracy.
- **With Investment in Labeling and Computing**: Expected to achieve **75-80%+** accuracy and own **strategic gold-standard data**, with:
  - VPBank DCD providing robust data support.
  - AWS contributing additional computing resources.
- **Investment Rationale**: Leverage AWS credits, DCD collaboration, and AIC expertise for a sustainable Callbot ecosystem.
- **VPBank's Vision**: As a top-tier bank in Vietnam, VPBank leads in AI-driven banking, driving digital transformation and delivering innovative financial services.
- **Next Steps**: Align with AWS, DCD, and AIC for additional resources to advance ASR and related applications.

Acknowledgments

- **To AWS**: Heartfelt thanks for the $14,000 GPU credits, enabling cutting-edge AI model development.
- **To DCD**: Gratitude for operational insights and collaboration, aligning AI solutions with debt collection needs.
- **To AIC**: Appreciation for relentless innovation and dedication to building world-class AI capabilities.

- **Our Commitment**: VPBank will deepen partnerships with AWS and DCD, leveraging AIC's expertise to drive AI innovation, enhance customer experiences, and maintain our position as a technology leader in banking.
- **Future Promise**: Continue delivering value to stakeholders, pushing the boundaries of AI-driven financial services, and fostering sustainable growth.

Contact

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